Poster 2-031

POTENTIATED PROCESSING OF HIGH-AROUSING TARGET STIMULI AS REVEALED BY EEG FREQUENCY ANALYSIS

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Descriptors: emotion,attention, α-ERD
Previous research revealed that explicit task relevance and implicit emotional stimulus significance are associated with alpha-band desynchronization (α-ERD). The main aim of the present study was to compare the α-ERD of explicit task and implicit emotional stimulus significance in terms of topography and frequency and to assess the interaction of target and emotion effects by comparing task effects to pleasant stimuli differing in emotional arousal. Towards this end, participants ($N=16$) viewed pictures from two stimulus categories. The 'undressed' category comprised erotic images and the 'dressed' category consistent of couples in romantic poses. In alternating blocks, participants were asked to respond either to the dressed or undressed picture category. Frequency was analyzed by a frequency transformation of the one second post-stimulus interval. Results indicate that the processing of stimuli from the 'undressed' as compared to 'dressed' picture category was associated with a reduced α-power over anterior and posterior clusters. Similar to emotion effects, reduced α-power was observed when contrasting target with non-target stimulus processing over anterior and posterior sensor regions. Of main interest, over posterior regions, a significant interaction of task by emotion was observed indicating stronger α-ERD for the 'undressed' compared to 'dressed' target stimuli. The present study replicated previous findings on the modulation of α-ERD by explicit and implicit stimulus significance and provided evidence for the potentiated processing of high-arousing target stimuli.

Poster 2-032

LINKS BETWEEN EMPATHIC SKILLS AND EXECUTIVE CONTROL IN DEPRESSION: THE CONTRIBUTION OF PHYSIOLOGICAL MEASURES

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Descriptors: depression, empathy, executive functions
Empathy consists of ability to infer and share others' emotional states. Cognitive theories posit that depression could be characterized by reduced empathy and executive control. The purpose of this study is to investigate emotional activation during empathy task in link with executive functioning. To this aim, we recruited subjects with a low or high level of depression on Beck Depression Inventory (BDI), and patients who received a diagnosis of major depressive disorder regarding the Hamilton's depression interview (HDRS) conducted by a psychologist/psychiatrist. To assess affective (AE) and cognitive empathy (CE), participants performed the Multifaceted Empathy Test (MET), which consists of photographs depicting people in emotionally charged situations. Subjects had to rate their level of arousal, the protagonist's emotional state, its valence and their level of compassion and distress. During the task, cardiac variability (CV), respiratory frequency (RSA), electro-dermal activity (EAD) and facial activation (EMG) were recorded. Subjects completed three Miyake's tasks (2000) measuring executive functions in neutral (rectangles) and emotional contexts (faces). Subjective empathy was assessed with the Basic Empathy Scale (BES). Results will be presented and discussed accordingly. Performances at the MET should reveal less empathy correlated with increased levels of depression and decreased executive control. Higher levels of depression should be linked with globally reduced CV and RSA, an enhanced EDA for sad faces and a reduced facial EMG for happy ones.

Poster 2-033

THE EFFECTS OF AMBIVALENCE AND EMOTION REPORTING ON ATTENTION

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Descriptors: ambivalence, attention, N200
Ambivalence, the co-occurrence of positive and negative affect, is uncomfortable, arousing, and short-lived. We have previously shown that ambivalence distracts individuals from performing subsequent tasks . Individuals experienced greater interference on a color Stroop task after watching an ambivalent vs. a positive or negative film clip; however, allowing individuals to report their feelings eliminated this impairment (Norris & Henderson, 2013). In the current study we sought to examine the effects of ambivalence and emotion reporting on attention, using the Attention Network Test (ANT; Fan et al., 2002) and event-related brain potentials (ERPs). Participants randomly assigned to report their feelings after watching an emotion-inducing film clip showed both better accuracy and longer response times on the ANT than non-reporters. These effects were driven by participants who viewed an ambivalent (vs. a negative or positive) film clip, and were strongest for incongruent (vs. neutral or congruent) ANT trials. N200 amplitudes were smaller for incongruent than for neutral or congruent trials; and were reduced for participants who watched the ambivalent film clip and reported their feelings afterward. Higher mixed feelings correlated with worse accuracy on the ANT and with smaller N200 amplitudes; this effect was driven by participants who viewed the ambivalent film clip. In sum, ambivalence may lead to impaired executive attention, as evidenced by both ANT accuracy and reduced N200 amplitudes, but reporting those mixed emotions may help alleviate these impairments.

Poster 2-034

THE NEURAL SIGNATURE OF VIEWING DYNAMIC EMOTIONAL FACES: COMPLEMENTARY CONTRIBUTIONS OF EVOKED VERSUS INDUCED OSCILLATORY BRAIN ACTIVITY

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Descriptors: emotion/affect, attention, EEG
The processing of facial expressions is often studied using static photographs. A body of recent research however suggests that dynamic expressions are more robust in evoking physiological responses in the observer, owing to their ecological validity. Here, we used sequences of neutral-to-affective and neutral-to-neutral facial expressions while recording electrocortical activity. Twenty-two participants viewed sequences of grayscale faces periodically turned on and off at a rate of 17.5 Hz, to evoke steady-state visual evoked potentials (ssVEPs). Each sequence began with a neutral face (flickering for 2290 ms), immediately followed by a flickering face from the same actor (also flickering for 2290 ms), with one of four expressions (happy, angry, fearful, or another neutral expression), followed by the initially presented neutral face (flickering for 1140 ms). Evoked activity (Hilbert transformation of the time-domain averages) and induced (intrinsic) brain oscillations (wavelet transform of each trial) were analyzed. We found a transient perturbation (reduction) of the ssVEP evoked by the face stream only in response to the neutral-to-angry change, selectively at right posterior sensors. Induced alpha-band power was reduced at mid-occipital sites, only after the first and after the second neutral-to-neutral change. Thus, the ssVEP showed modulation consistent with involvement of face-sensitive cortical areas in affective expression decoding, whereas alpha-band changes varied with perceptual similarity and trial statistics, consistent with its role in selective attention.

Poster 2-035

SPONTANEOUS AND MOTIVATION-ELICITED PROACTIVE CONTROL OF EMOTIONAL PROCESSING

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Descriptors: motivation, emotion, proactive control
Attention to emotional content is prioritized, but can be modulated by top-down control. The current study uses pupil size to index cognitive control and emotional arousal to examine spontaneous (Study 1) and motivation-elicited (Study 2) proactive control of emotional processing. In Study 1, participants passively viewed briefly-presented intact images that were interspersed with phase-scrambled images. The positive, negative, and neutral images were blocked by valence to create an emotional context; positive and negative images were matched on arousal. In the pre-stimulus period, phasic pupil dilation was modulated by emotional context, showing greater preparation for an upcoming image in negative than positive blocks. In contrast, greater tonic pupil size was found in positive compared to negative blocks, indicating higher sustained arousal. Intact positive images also elicited greater phasic post-image pupil dilation compared to negative ones, replicating previous work (e.g. Henderson, Bradley, & Lang, 2014). Findings suggest greater spontaneous preparation for negative images; and greater phasic and tonic arousal in a positive emotional context. In a preregistered second study, we use pupillometry to determine whether motivation (induced by performance-contingent financial reward) elicits a shift to anticipatory proactive control, and reduced processing of the emotional images.